

Changes in nerve growth factor in vastus lateralis muscle after the first versus second bout of one-leg eccentric cycling

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Abstract

Delayed onset muscle soreness (DOMS) develops after performing unaccustomed eccentric exercises. Animal studies have shown that DOMS is mechanical hyperalgesia through nociceptor sensitization induced by nerve growth factor (NGF) and glial cell line-derived neurotrophic factor (GDNF) upregulated by cyclooxygenase-2 (COX-2). However, no previous study has investigated these in relation to DOMS in humans. This study compared the first and second bouts of one-leg eccentric cycling (ECC) for changes in NGF, GDNF, and COX-2 mRNA in the vastus lateralis (VL). Seven healthy adults (18–40 years) performed two bouts of ECC (10 sets of 50 contractions) with 80% maximal voluntary concentric peak torque separated by 2 weeks (ECC1, ECC2). Muscle soreness that was assessed by a visual analog scale and maximal voluntary isometric contraction (MVC) torque of the knee extensors were measured before, immediately after (MVC only), 24 and 48 h post-exercise. Muscle biopsy was taken from the VL before the first bout from nonexercised leg (control) and 24 h after each bout from the exercised leg, and analyzed for NGF, GDNF, and COX-2 mRNA. Peak DOMS was more than two times greater and MVC torque at 48 h post-exercise was approximately 20% smaller after ECC1 than ECC2 ($p < 0.05$), suggesting the repeated bout effect. NGF mRNA level was higher ($p < 0.05$) post-ECC1 (0.79 ± 0.68 arbitrary unit) than control (0.06 ± 0.07) and post-ECC2 (0.08 ± 0.10). GDNF and COX-2 mRNA did not show significant differences between control, post-ECC1, and post-ECC2. These results suggest that an increase in NGF is associated with the development of DOMS in humans.

KEYWORDS

cyclooxygenase-2, delayed onset muscle soreness, glial cell line-derived neurotrophic factor, mechanical hyperalgesia, muscle damage, repeated bout effect

1 | INTRODUCTION

Delayed onset muscle soreness (DOMS) is a movement-related pain and increased tenderness, peaks at 24–72 h, and normally disappears within 5–7 days after unaccustomed exercise.^{1–3} Delayed onset muscle soreness has been considered to reflect muscle damage induced by eccentric exercise,^{3–8} and following inflammation of muscle fibers^{2,7,8} and connective tissue surrounding muscle fibers, fascicles, and fascia.^{3,9} Sonkodi et al.¹⁰ have recently hypothesized that DOMS is an acute compression axonopathy of the nerve endings in the muscle spindle, which could coincide with micro-injury of the surrounding tissues, but solid experimental evidence supporting this hypothesis is lacking. Thus, the whole picture concerning the mechanisms underpinning DOMS is still not clear.

Animal studies have shown that eccentric muscle contractions induce mechanical hyperalgesia in the exercised muscle and increase c-Fos expression in the spinal dorsal horn after compression of the exercised muscle.¹¹ Hayashi et al.¹² reported that mechanical hyperalgesia occurred in a stretch velocity-dependent and range of motion (ROM)-dependent manner in eccentric muscle contractions of rat dorsiflexor muscles. They showed that the greater the rate of torque increase and integrated torque quantity, the greater the magnitude of mechanical hyperalgesia.¹² They also observed no muscle fiber damage such as necrotic myofibers infiltrated by mononuclear cells and myofibers with central nuclei except for the highest velocity and largest ROM in some animals, concluding that muscle fiber damage at the light microscopic level was not essential for the induction of mechanical hyperalgesia (DOMS).¹²

Animal studies have found that nerve growth factor (NGF), a neurotrophic factor known to induce hyperalgesia,¹³ is upregulated through bradykinin B2 receptor activation during eccentric exercise,¹⁴ and glial cell line-derived neurotrophic factor (GDNF) is upregulated through activation of cyclooxygenase (COX)-2, resulting in mechanical hyperalgesia.^{15,16} NGF and GDNF are upregulated at 12 h, 1, and 2 days after eccentric muscle contractions in the satellite cells and/or myofibers, but not in inflammatory cells,^{14,15,17} sensitizing C-fibers¹⁴ and A δ -fibers,¹⁶ respectively. The reversal of mechanical hyperalgesia by intramuscular injection of anti-NGF or anti-GDNF antibodies has confirmed a causal relationship between these neurotrophic factors and mechanical hyperalgesia after eccentric muscle contractions.^{14,15} These molecular and peripheral neural mechanisms explain why mechanical hyperalgesia (DOMS) occurs with some delay after eccentric exercise in animals.¹⁸ Andersen

et al.¹⁹ reported injection into the tibialis anterior of NGF caused hyperalgesia at the injection site 3 h after injection and increased muscle soreness for up to 7 days in humans. However, it has not been investigated whether NGF is involved in human DOMS.

Repeating the same eccentric exercise within several weeks induces less DOMS and faster recovery of muscle function when compared with the initial bout, which is referred to as the repeated bout effect (RBE).^{1,20} The RBE has also been observed in the rat model such that Urai et al.²¹ reported that NGF upregulation was blunted in the second bout of eccentric exercise. However, no previous study has examined changes in NGF, GDNF, and COX-2 in relation to DOMS and the RBE in humans.

Therefore, we examined NGF, GDNF, and COX-2 mRNA levels in exercised muscles, as well as muscle soreness and muscle strength after two bouts of eccentric cycling with one leg separated by two weeks in humans. We hypothesized that the NGF, GDNF, and COX-2 mRNA levels would increase after the first exercise bout, but the magnitude of increases would be smaller after the second than the first bout of the one-leg eccentric cycling that was performed by the same leg.

2 | MATERIALS AND METHODS

2.1 | Participants

This study was approved by the institution's human ethics committee (project number: 19767). We initially recruited 11 participants for this study, but two participants dropped out after the first exercise bout, and we were unable to detect NGF mRNA in the control samples of two participants who completed two bouts of one-leg eccentric cycling. Thus, the participants included in this study were seven healthy volunteers (two men and five women) aged 18–40 (mean $\pm$ SD; 31.0 $\pm$ 8.1) years. Their mean $\pm$ SD height was 164.5 $\pm$ 6.3 cm, body mass was 62.6 $\pm$ 10.5 kg, and body mass index was 23.1 $\pm$ 3.1 kg/m². Participants had never performed one-leg eccentric cycling before the present study and did not perform any regular resistance exercise of lower extremities in the last 6 months before participating in the study. They were free from musculoskeletal or neurological disorders. Each participant signed a written informed consent form after receiving full information regarding the risks and benefits of this study and provided answers to medical questionnaires before participating in this study. They were asked not to perform any exercise during the experimental period and to have a similar diet for 24 h before the first and second exercise bouts.

2.2 | One-leg eccentric cycling

Participants performed two bouts of one-leg eccentric cycling (ECC1 and ECC2) with their right legs using a recumbent cycle ergometer (Strength Ergo 240, Mitsubishi Electric Engineering, Tokyo, Japan). Each participant sat on the seat of the ergometer and was instructed to perform backward rotational movements of the crank without resistance for 2 min. After confirming that they were ready to perform backward cycling, the participants were instructed to resist the pedal rotating backward at 50 rpm with the target torque at 80% of maximal torque in 10-s maximal concentric cycling (Max-torque). The exercise consisted of 10 sets of 1-min eccentric cycling with a 1-min break between sets. During the exercise, the target torque was displayed on a computer screen monitor, and the participants were instructed to match the force output of each revolution to the target torque as much as possible. An investigator counted the number of revolutions that reached the target torque up to 50 times for each set.

2.3 | Familiarization session

A couple of days before the first eccentric cycling (ECC1), each participant had a familiarization session to become acquainted with the measurements of muscle soreness, maximum voluntary isometric, and isokinetic contraction torque of the knee extensors and flexors, and the 10-s maximal concentric cycling to measure the Max-torque in this order. No eccentric cycling was performed in the familiarization session to avoid the repeated bout effect.

2.4 | Muscle soreness

Each participant lay on a massage table and was asked to mark the level of perceived soreness on a 100-mm visual analog scale (VAS), where 0 represents no soreness and 100 as “intolerable pain”²² when the vastus lateralis (VL) and rectus femoris (RF) were palpated by the investigator before, 24 and 48 h after each bout. The investigator had been trained to apply constant pressure, which was approximately 2 kg force to the muscles before the study. Three sites over the VL (25%, 50%, and 75% points between the apex patella and greater trochanter) and the RF (the same 25%, 50%, and 75% points projected from the sites of VL) were marked with a semipermanent ink pen. The average of three measurement sites for each muscle was used for further analysis. Movement-induced muscle soreness of the knee extensors was also assessed by the

VAS during a squat in which each participant slowly stood up and sat down on a 42-cm chair in height.²³ Peak muscle soreness was considered as the higher VAS between the 24- and 48-h post-exercise values, and the difference to the baseline value (i.e., peak VAS – baseline VAS) was used for further analyses.

2.5 | Maximum voluntary isometric contraction (MVCiso) and isokinetic concentric contraction (MVCcon) torque

MVCiso torque of the right knee extensors and flexors was measured at 20° knee flexion (0° = full extension of the knee joint) after several submaximal contractions as a warm-up²⁴ using an isokinetic dynamometer (BIODEX System 3, Biodex Medical Systems, USA) connected to a digital data acquisition system (PowerLab, ADInstruments Pty Ltd, Australia). A total of three 3-s maximum contractions were performed, separated by a 1-min rest between contractions. Dynamic MVC torque (MVCcon torque) of the knee extensors and flexors was also measured using the isokinetic dynamometer at an angular velocity of 90°/s. The participants maximally extended their knee from 90° to 20° and then flexed to 90° as fast and hard as possible for four consecutive contractions. The highest torque value in each direction was used for further analysis. These measurements were taken before (pre), immediately after, 24 and 48 h after (post) ECC1 and ECC2. The measurement at 24 h after ECC was taken after the muscle biopsy, which might be affected by the biopsy. Thus, we used the MVC torque measures at 48-h post-exercise normalized to the pre-exercise values to compare between the first and second bouts.

2.6 | NGF, GDNF, and COX-2 mRNA and NGF protein measurements

Muscle samples were taken from the nonexercised (left) VL before ECC1 and taken from the exercised (right) VL at 24 h after ECC1 and ECC2 using a microbiopsy technique. The first and second biopsy samples were taken at half the distance between the lateral epicondyle and greater trochanter in the left and right VL, respectively. The third one was taken approximately 2 cm distal from the previous needle insertion site to avoid the tissue possibly affected by the previous biopsy. The skin at the biopsy site was anesthetized with cream containing 2.5% lignocaine and 2.5% prilocaine (Emla cream, Aspen Pharmacare Australia Pty Ltd, Australia), and the cream was removed 30 min later. A needle of the biopsy system (MAX = CORE, Bard Biopsy System, USA) was inserted into the mid-belly

of the VL up to a depth of approximately 2 cm, a biopsy gun was inserted three times into the hole made by the needle, and a muscle sample was taken. Approximately 15 mg of muscle tissue was obtained each time, and a total amount of 40–50 mg was obtained in three attempts from the same insertion site. One-third of each sample was immediately immersed in RNAlater solution (Thermo Fisher Scientific) at room temperature, stored overnight at 4°C, and then stored at –80°C until further analysis. The rest (two-thirds of each sample) was immediately immersed in liquid nitrogen to freeze, placed in a tube, and stored at –80°C until further analyses.

Tissue samples were homogenized and total RNAs were extracted using the RNeasy Fibrous Tissue Mini Kit (QIAGEN, Germany) according to the manufacturer's instructions. The complementary DNA was prepared with High-Capacity cDNA Reverse Transcription Kit (Thermo Fisher Scientific) using random primers. Quantitative polymerase chain reaction (qPCR) was performed with the THUNDERBIRD SYBR qPCR Mix (Toyobo Co., Ltd, Japan) and the gene-specific primer sets (QuantiTect Primer Assays, QIAGEN) using Applied Biosystems 7500 Real-Time PCR System (Thermo Fisher Scientific). A three-step thermal protocol, 40 cycles of 95°C for 3 s, 55°C for 5 s, and 72°C for 35 s, was used for amplification and followed by a melting curve analysis to assess the specificities of the products. Relative gene expression levels were determined with the standard curve methods and normalized to the transcript levels of glyceraldehyde-3-phosphate dehydrogenase (GAPDH) used as an internal control. Two-fold serial dilutions of the human muscle cDNA were used as the standards. All the samples and standards were duplicated and averaged.

Nerve growth factor protein in the muscle tissue extract was measured by an enzyme-linked immunosorbent assay (ELISA) using a kit (KA0399, Abnova, Taiwan) according to the manufacturer's instructions. The minimum assay sensitivity of NGF was 7.8 pg/mL. All measurements were taken in duplicate, and the amount of NGF protein per 1 mg of tissue protein was calculated.

2.7 | Statistical analyses

The normality of the data was assessed using the Shapiro–Wilk test. We analyzed the muscle soreness (VAS) with palpation, MVCiso, and MVCcon data using a parametric paired *t*-test but used a nonparametric Wilcoxon rank test for the muscle soreness (VAS) during squat. Friedman test with multiple comparisons (Dunn's test) was used to compare mRNA levels of GAPDH, NGF, GDNF, and COX-2 and NGF protein. The significance level was set at

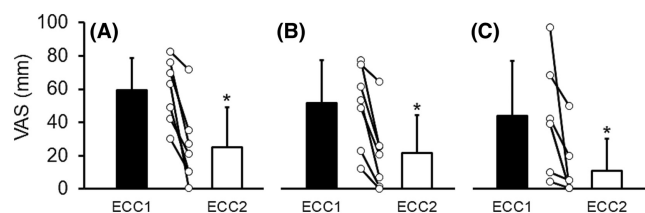


FIGURE 1 Peak muscle soreness assessed by a visual analog scale (VAS) for palpation of vastus lateralis (A) and rectus femoris (B), and during a squat movement (C) for individuals as well as the average and standard deviation of seven participants after the first (ECC1) and the second bout of one-leg eccentric cycling (ECC2).

*: a significant ($p < 0.05$) difference between ECC1 and ECC2 by a paired *t*-test or a Wilcoxon's rank test.

$p < 0.05$. All statistical analyses were performed using R (version 2.8.1) and GraphPad Prism (version 9). Data are presented as mean $\pm$ standard deviation (SD).

3 | RESULTS

3.1 | Eccentric cycling

All participants performed two bouts of eccentric cycling as planned. Max-torque measured in the 10-s maximal concentric cycling to set the 80% target in eccentric cycling was similar ($p = 0.97$) between ECC1 (126.7 ± 9.0 Nm; range: 84.5–159.1 Nm) and ECC2 (129.7 ± 10.5 Nm; 85.8–177.0 Nm). The total work during the eccentric cycling represented by the sum of the torques produced in 500 eccentric contractions was also similar ($p = 0.12$) between ECC1 ($68\,343 \pm 11\,219$ Nm) and ECC2 ($66\,092 \pm 10\,319$ Nm).

3.2 | Muscle soreness

The pre-exercise VAS values were less than 10 mm for all measures, and no significant differences were evident between ECC1 and ECC2 (VL; $p = 0.58$, RF; $p = 0.84$, squat; $p = 0.81$). Muscle soreness (VAS) increased at 24 h (54.6 ± 21.4 mm) and 48 h after ECC1 (65.8 ± 19.9 mm) with VL palpation, and this was also the case for the RF palpation (24 h: 45.7 ± 26.1 mm, 48 h: 54.4 ± 30.3 mm) and during squat (24 h: 30.0 ± 33.3 mm, 48 h: 43.9 ± 33.2 mm). These increases were smaller ($p = 0.01$ – 0.04 for 24 h and 48 h) after ECC2 than ECC1. As shown in Figure 1, all participants had less peak soreness after ECC2 than ECC1 (less than 50% of ECC1 after ECC2), and a significant difference between bouts was evident for palpation of VL (Figure 1A, $p = 0.003$) and RF (Figure 1B, $p = 0.003$), and during squat (Figure 1C, $p = 0.022$).

3.3 | MVCiso and MVCcon torque

The pre-exercise MVCiso and MVCcon torque of knee extensors was not significantly different between bouts (1st bout: 77.3 ± 7.5 Nm, 2nd bout 74.8 ± 8.3 Nm, $p = 0.90$ and 128.5 ± 16.5 Nm, 119.6 ± 17.6 Nm, $p = 0.35$, respectively). Both MVCiso and MVCcon torques decreased from the baseline (pre-exercise, before ECC1) values immediately after ECC1 (58.2 ± 11.6 Nm and 104.1 ± 23.3 Nm, respectively), and still below the baseline values at 48 h after (59.0 ± 10.1 Nm and 91.3 ± 25.6 Nm, respectively). Both MVC measures returned to the baseline before ECC2 (as described above) but decreased immediately after ECC2 (65.1 ± 8.7 Nm and 111.6 ± 14.8 Nm, respectively) again, however, recovered faster after ECC2 than ECC1. Some participants were not able to generate maximal force output due to pain after muscle biopsy at 24 h post-exercise. Thus, it is possible that the MVC values at 24 h after exercise did not represent the effects of exercise alone. Based on these, we used the 48-h post-exercise measures to compare between bouts. As shown in Figure 2, all participants showed higher MVC torque values at 48 h after ECC2 than ECC1, indicating that the difference from the pre-exercise value was smaller after ECC2 (MVCiso: $-2.5 \pm 15.5\%$, MVCcon: $-11.1 \pm 18.6\%$) than ECC1 (MVCiso: $-23.9 \pm 10.1\%$, MVCcon: $-29.2 \pm 16.3\%$) for both MVCiso ($p = 0.013$) and MVCcon ($p = 0.023$). Regarding the knee flexors, no significant differences in the pre-exercise values between bouts, and no significant changes were observed for MVCiso ($p = 0.31$) and MVCcon torque ($p = 0.17$) after ECC1 and ECC2 (data not shown).

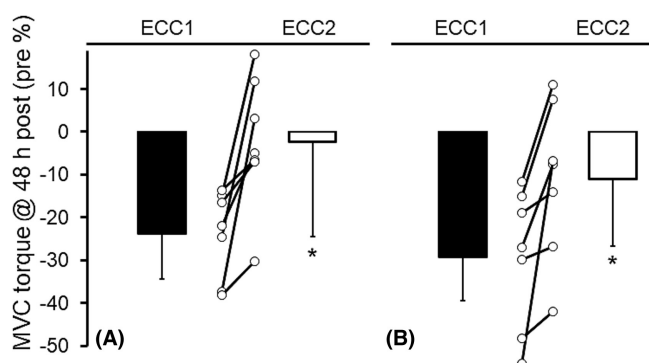


FIGURE 2 Maximal voluntary isometric contraction torque of the knee extensors at 20° knee flexion (MVCiso, A) and maximal voluntary isokinetic concentric contraction torque of the knee extensors (MVCcon, B) at 48 h after exercise normalized to pre-exercise values. Data from the first (ECC1) and the second bout of one-leg eccentric cycling (ECC2) are presented for individuals comparing between bouts as well as the average and standard deviation of seven participants in each bout. *a significant ($p < 0.05$) difference between ECC1 and ECC2 by a paired t-test.

3.4 | NGF, GDNF, and COX-2 mRNA and NGF protein

The values of GAPDH mRNA were similar ($p = 0.65$) between the control (7.42 ± 3.56), post-ECC1 (7.46 ± 4.95), and post-ECC2 (6.92 ± 3.06). NGF mRNA levels (relative to internal standard GAPDH mRNA) in the control samples were near the detection limit. As shown in Figure 3, NGF mRNA was significantly higher ($p = 0.001$, the effect size was 1.13) at 24 h after ECC1 than the control ($p = 0.049$), and the level after ECC2 was similar to the control ($p = 0.99$) and lower than that after ECC1 ($p = 0.004$). The mRNAs levels of GDNF and COX-2 relative to GAPDH mRNA were not significantly different between post-ECC1 and post-ECC2 ($p = 0.56$ and $p = 0.65$, respectively).

Nerve growth factor protein levels were not significantly different ($p = 0.37$) between the control (108.4 ± 31.8 pg/mg), post-ECC1 (107.0 ± 25.4 pg/mg), and post-ECC2 (96.2 ± 29.2 pg/mg).

4 | DISCUSSION

To the best of our knowledge, this was the first study to investigate NGF, GDNF, and COX-2 mRNA after eccentric exercise (i.e., one-leg eccentric cycling) in relation to DOMS in humans. The less muscle soreness and faster recovery in MVC torque after ECC2 than ECC1 indicate the repeated bout effect. NGF mRNA level was higher post-ECC1 than the baseline (control) and post-ECC2 (no significant difference from the control), but no significant differences were observed between control, post-ECC1, and post-ECC2 for GDNF and COX-2 mRNAs. We did not perform muscle biopsy before ECC2 to minimize the number of biopsies as much as possible to reduce the burden to the participants. Since muscle soreness and MVC torque measures before ECC2 were not significantly different from those before ECC1, it was assumed that the increased NGF mRNA after ECC1 returned to the control level before ECC2. It is important to note that NGF mRNA level after ECC2 was not different from that of the control. Thus, it is likely that NGF mRNA was greater after ECC1 than ECC2 and that the mRNA did not increase after ECC2. These results supported the hypothesis that an increase in NGF would be associated with DOMS.

We used GAPDH mRNA as an internal standard, according to our previous animal studies in which mRNAs of NGF, GDNF, and COX2 were investigated in response to eccentric exercise.^{14,15,21} In the present study, we found that GAPDH mRNA level was constant between the three samples (control, post-ECC1, post-ECC2). Therefore, we think that the use of GAPDH as an internal standard was valid. We measured NGF protein, but it did not show any

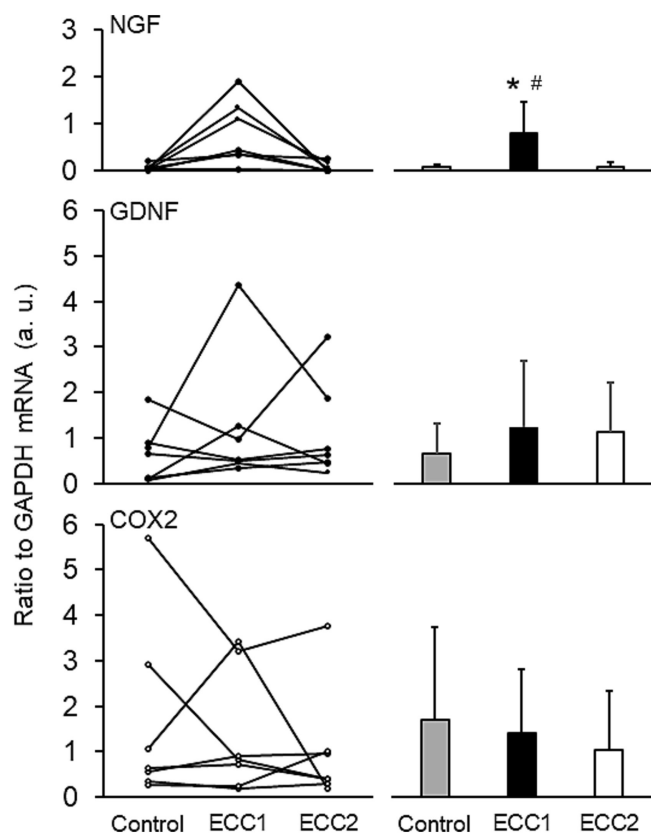


FIGURE 3 mRNA levels of nerve growth factor (NGF), glial cell line-derived neurotrophic factor (GDNF), and cyclooxygenase (COX)-2 of the vastus lateralis muscle taken before the first one-leg eccentric exercise bout from the non-exercised leg (control) and at 24 h after the first (ECC1) and second bout (ECC2) from the exercised leg. The average and standard deviation of seven participants (right) and individual changes (left) are shown.

* $p < 0.05$ vs. Control, # $p < 0.05$ vs ECC2 based on Dunn's test.

significant difference between control, post-ECC1, and post-ECC2 in the present study. The relatively high concentration of NGF protein in the control sample in comparison with the low level of NGF mRNA may be due to the nonspecific binding of some muscle proteins to the antibody used in the ELISA. Another possibility is that pro-NGF was also measured, and the total amount of pro-NGF and NGF was not influenced by exercise.

The present study used one-leg eccentric cycling in which 500 eccentric contractions of the knee extensors were performed. Since eccentric cycling was an unaccustomed exercise for all participants, it was expected that it would induce muscle damage, especially after the first bout.^{2,3} Indeed, muscle soreness appeared at 24 h and lasted up to 48 h after ECC1, and MVC torque decreased immediately after ECC1 and did not recover at 48 h after ECC1. We did not collect blood or urine samples to assess muscle damage (e.g., creatine kinase activity) in the present study and did not examine circulating neutrophils and other inflammation markers in the blood that had been reported

to increase after eccentric exercise.^{2,7,8,10} Thus, the whole picture of muscle damage is not clear in the present study. However, the two exercise bouts were performed similarly as indicated by the similar total torque production during the cycling, and despite this, muscle soreness was smaller and the decrease of MVC torque was smaller after ECC2 than ECC1 (Figures 1 and 2). It seems likely that changes in blood markers of muscle damage and inflammation were also smaller after ECC2 than ECC1. The less magnitude of muscle damage after ECC2 than ECC1 represents the repeated bout effect.^{2,3}

Peñailillo et al.¹ investigated muscle damage responses to two eccentric cycling bouts (30 min at 60% of the maximal concentric power output at 60 rpm) separated by two weeks. They reported that decreases in MVC strength of the knee extensors (approximately 15% at 1 day post-exercise) and increases in muscle soreness (approximately 20 mm in a 100-mm VAS scale at 1 day post-exercise) were observed only after the first bout. Although the eccentric cycling protocol was different between the study by Peñailillo et al.¹ (continuous 30 min cycling with both legs at 60 rpm in 60% of the maximal concentric power output) and the present study (10 sets of 1-min intermittent cycling with one leg at 50 rpm in 80% of maximal concentric cycling load), changes in the MVC strength and muscle soreness following the first eccentric cycling bout were comparable. When comparing ECC1 and ECC2, peak muscle soreness with palpation of the VL and RF and during squat were higher after ECC1 than ECC2 (Figure 1), and MVCiso and MVCcon torque of the knee extensors at 48-h post-exercise were smaller after ECC1 than ECC2 (Figure 2). These were also in line with the repeated bout effect reported in the previous studies.^{1,23,24} Thus, this was ideal for investigating whether the upregulation of NGF, GDNF, and COX-2 in VL would be associated with DOMS.

Murase et al.^{14,15} have shown in animal studies that NGF and GDNF were causal agents (sensitizers for nociceptors) for mechanical hyperalgesia (DOMS). They reported that upregulation of GDNF at 12 h and 1 day after 500 eccentric contractions of extensor digitorum longus (EDL) was induced by upregulation of COX-2 (0–12 h), whereas NGF upregulation occurred at 12 h–2 days after the eccentric exercise through the activation of B2 bradykinin receptors.¹⁴ Injection of anti-GDNF and anti-NGF antibodies to the exercised muscle reversed the mechanical hyperalgesia; thus, these neurotrophic factors were considered to be causal agents of the mechanical hyperalgesia.^{14,15} If these are also the cause for DOMS in humans, it was assumed that NGF and GDNF mRNAs would be upregulated, but COX-2 mRNA might not change at 24 h post-ECC1. NGF mRNA level at 24-h post-ECC1 was significantly higher than that of control, and the level after ECC2 was not different from the control (Figure 3), indicating no increase

after ECC2. This suggests that NGF upregulation is likely to be involved in DOMS in humans. Urai et al.²¹ reported that NGF upregulation was blunted after the second bout of eccentric exercise consisting of 500 eccentric contractions of EDL performed 5 days after the first bout, together with an absence of mechanical hyperalgesia in rats. The present findings were in line with their study.

COX-2 mRNA upregulation was observed between 0 and 12 h after eccentric exercise of EDL in the animal model,¹⁵ and GDNF mRNA upregulation was observed at 12 h – 1 day after the eccentric exercise. It should be noted that the muscle samples were taken 24 h after the eccentric cycling in the present study. This might be the reason why we did not observe increases in COX-2 and GDNF mRNAs after eccentric cycling from the baseline. The timing of biopsy might have not been the best for detecting changes in GDNF mRNA and COX-2 mRNA. As shown in Figure 3, two participants showed an increase in GDNF mRNA after ECC1 only, but this was not the case for COX-2 mRNA. Performing several biopsies was not possible in the present study; therefore, we limited biopsies to two times from the exercised leg. If we had been able to take more biopsy samples at an earlier time point after the exercise (e.g., 12-h post-exercise), upregulation of mRNAs of COX-2 and GDNF might have been observed as reported by Murase et al.¹⁵ Additionally, we observed large variability among the participants for GDNF and COX2 mRNA as shown in Figure 3. Therefore, we need to recruit more participants to confirm the results.

Nerve growth factor is not the sole molecule that is involved in DOMS, and the cause–effect relationship between NGF and DOMS could not be determined from the present study. We have shown in our previous animal study^{25,26} that transient receptor potential vanilloid 1 (TRPV1) and transient receptor potential vanilloid 4 (TRPV4) are involved in the upstream and/or downstream of GDNF and NGF, and prostaglandin E2 (EP2) receptor is involved in the upstream of GDNF production. Before identifying these in human studies, we thought that the first step was to investigate changes in NGF and GDNF in relation to DOMS after the first and second eccentric exercise bouts. However, TRPV1, TRPV4, and EP2 should be investigated in future human studies.

The present study has several limitations. First, the sample size was small. We initially recruited 11 participants, but two participants dropped out after the first exercise bout due to pain after the biopsy. We had to exclude two more participants since the NGF mRNAs were undetectable in their samples. Thus, the number of participants included in the present study was seven, and no correlation analyses were possible. We compared peak muscle soreness between bouts and found that the effect size was large ($d = 1 \sim 1.86$). We also found that the effect size for

the difference in MVC torque at 48-h post-exercise relative to the pre-exercise level was also large between bouts ($d = 1.14 \sim 1.32$). Thus, the present sample size appeared to be acceptable for the muscle soreness and MVC measures, showing the typical repeated bout effect, and importantly DOMS after the second bout was minor. We also obtained a large effect size (1.13) between post-ECC1 and post-ECC2 for the NGF mRNA level. Thus, we are confident about the increase in NGF mRNA only after the first bout. Second, the muscle sample was taken only three time points from each participant, the first was taken from the non-exercising leg before the exercise, and two samples were from the exercised leg 24 h after each bout. We assumed that NGF mRNA level before ECC2 would be similar to the level before ECC1. As discussed above, animal studies¹⁵ showed that GDNF mRNA increased at 12 h and 1 day after eccentric exercise, and COX-2 mRNA increased immediately after and 12 h after the exercise. Therefore, no changes in GDNF mRNA and COX-2 mRNA might be due to the timing of the biopsy in the present study. Additionally, due to the small amount of muscle biopsy samples, we did not measure GDNF and COX2 proteins. Thus, the findings of the present study should be confirmed by a larger sample size with more biopsy samples in the time course. Third, we could not clarify the causal relationship between NGF/GDNF and DOMS in the present study. To do this, anti-NGF/GDNF antibodies that could be injected into the human body to neutralize NGF and GDNF in the muscle are necessary, as we had done in our previous animal study.^{14,15} This could not be done due to the nonavailability of such antibodies, and no ethics approval for this procedure. Despite the limitations, it is important that the present study showed that NGF was a molecule responsible for DOMS in humans. It is interesting to investigate how NGF is produced (e.g., muscle cells, satellite cells, and inflammatory cells) in response to eccentric exercise in future studies.

In conclusion, the NGF mRNA level increased after the first one-leg eccentric cycling exercise that induced DOMS and decreases in muscle function, but no such increase was observed after the second one-leg eccentric cycling bout that was performed 2 weeks later resulting in minimal DOMS. This suggests that an increase in NGF is involved in DOMS in humans.

5 | PERSPECTIVE

To the best of our knowledge, this was the first study to use one-leg eccentric cycling for the investigation of muscle damage. Participants performed 10 sets of 1-min intermittent eccentric cycling at 50 rpm at 80% of maximal concentric cycling load. The exercise induced typical symptoms of

muscle damage (DOMS, a prolonged loss of MVC torque) after the first bout. This may be a good model for future muscle damage studies. One-leg eccentric cycling is also useful as an exercise intervention for patients who have an injury of one-leg since eccentric exercise training prevents muscle wasting.³ Although DOMS could be minimized by preconditioning exercise or gradually increasing the intensity and volume of eccentric exercise, many people are discouraged from performing regular exercise due to DOMS. Therefore, a strategy to reduce DOMS is still important. An anti-NGF substance is one of the candidates as an animal study¹⁴ reported that injection of anti-NGF antibody to the exercised muscle prevented and reversed DOMS. If such an effect can be produced by ingestion of a drug and it is proven to be safe and effective, it may be helpful for sedentary individuals to exercise regularly without DOMS.

AUTHORS CONTRIBUTION

TK, KM, and KN were involved in concept and study design, and manuscript preparation; TK, GM, and KN were involved in data collection; TK and KK were involved in muscle sample analyses; TK, KM, KK, and KN were involved in data analyses; All authors confirmed the final version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

Data are available from the corresponding author upon reasonable request.

PARTICIPANT CONSENT

An information letter was provided to each participant, and a written informed consent was obtained from each participant.

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REFERENCES

- Peñailillo L, Blazevich A, Numazawa NK. Metabolic and muscle damage profiles of concentric versus repeated eccentric cycling. *Med Sci Sports Exerc.* 2013;45:1773-1781.
- Kanda K, Sugama K, Hayashida H, et al. Eccentric exercise-induced delayed-onset muscle soreness and change in markers of muscle damage and inflammation. *Exerc Immunol Rev.* 2013;19:72-84.
- Hody S, Croiser JL, Bury T, et al. Eccentric muscle contractions: risk and benefits. *Front Physiol.* 2019;10:536.
- Friden J, Sjöström M, Ekblom B. Myofibrillar damage following intense eccentric exercise in man. *Int J Sports Med.* 1983;4:170-176.
- Newham DJ, McPhail G, Mills KR, Edwards RHT. Ultrastructural changes after concentric and eccentric contractions of human muscle. *J Neurol Sci.* 1983;61:109-122.
- Armstrong RB, Ogilvie RW, Schwane JA. Eccentric exercise-induced injury to rat skeletal muscle. *J Appl Physiol.* 1983;54:80-93.
- Paulsen G, Crameri R, Benestad HB, et al. Time course of leukocyte accumulation in human muscle after eccentric exercise. *Med Sci Sports Exerc.* 2010;42:75-85.
- Paulsen G, Mikkelsen UR, Raastad T, et al. Leukocytes, cytokines and satellite cells: what role do they play in muscle damage and regeneration following eccentric exercise? *Exerc Immunol Rev.* 2012;18:42-97.
- Wilke J, Behringer M. Is "delayed onset muscle soreness" a false friend? The potential implication of the fascial connective tissue in post-exercise discomfort. *Int J Mol Sci.* 2021;22:9482.
- Sonkodi B, Berkes I, Koltai E. Have we looked in the wrong direction for more than 100 years? Delayed onset muscle soreness is, in fact, neural microdamage rather than muscle damage. *Antioxidants.* 2020;9:212.
- Taguchi T, Matsuda T, Tamura R, Sato J, Mizumura K. Muscular mechanical hyperalgesia revealed by behavioural pain test and c-Fos expression in the spinal dorsal horn after eccentric contraction in rats. *J Physiol.* 2005;564:259-268.
- Hayashi K, Katanosaka K, Abe M, et al. Muscular mechanical hyperalgesia after lengthening contractions in rats depends on stretch velocity and range of motion. *Eur J Pain.* 2016;21:125-139.
- Mizumura K, Kubo A. Neurotrophins and pain. In: Frizsch B, Pogatzk-Zahn E, Schaible H-G, eds. *The Senses: A Comprehensive Reference*. Vol 5. Elsevier, Academic Press; 2020:142-162.
- Murase S, Terazawa E, Queme F, et al. Bradykinin and nerve growth factor play pivotal roles in muscular mechanical hyperalgesia after exercise (delayed-onset muscle soreness). *J Neurosci.* 2010;30:3752-3761.
- Murase S, Terazawa E, Hirate K, et al. Upregulated glial cell line-derived neurotrophic factor through cyclooxygenase-2 activation in the muscle is required for mechanical hyperalgesia after exercise in rats. *J Physiol.* 2013;591:3035-3048.

16. Murase S, Kato K, Taguchi T, Mizumura K. Glial cell line-derived neurotrophic factor sensitized the mechanical response of muscular thin-fibre afferents in rats. *Eur J Pain*. 2014;18:629-638.
17. Murase S, Yamanaka Y, Kanda H, et al. COX-2, nerve growth factor (NGF) and glial cell-derived neurotrophic factor (GDNF), which play pivotal roles in delayed onset muscle soreness (DOMS), are produced by exercised skeletal muscle. *J Physiol Sci*. 2012;62:S179.
18. Mizumura K, Taguchi T. Delayed onset muscle soreness: involvement of neurotrophic factors. *J Physiol Sci*. 2016;66:43-52.
19. Andersen H, Arendt-Nielsen L, Svensson P, et al. Spatial and temporal aspects of muscle hyperalgesia induced by nerve growth factor in humans. *Exp Brain Res*. 2008;191:371-382.
20. Nosaka K, Clarkson PM. Muscle damage following repeated bouts of high force eccentric exercise. *Med Sci Sports Exerc*. 1995;27:1263-1269.
21. Urai H, Murase S, Mizumura K. Decreased nerve growth factor upregulation is a mechanism for reduced mechanical hyperalgesia after the second bout of exercise in rats. *Scand J Med Sci Sports*. 2013;23:e96-e101.
22. Lau WY, Muthalib MM, Nosaka K. Visual analog scale and pressure pain threshold for delayed onset muscle soreness assessment. *J Musculoskeletal Pain*. 2013;21:320-326.
23. Peñailillo L, Blazeovich A, Nosaka K. Muscle fascicle behavior during eccentric cycling and its relation to muscle soreness. *Med Sci Sports Exerc*. 2015;47:708-717.
24. Mavropalias G, Koeda T, Barley OR, et al. Comparison between high- and low-intensity eccentric cycling of equal mechanical work for muscle damage and the repeated bout effect. *Eur J Appl Physiol*. 2020;120:1015-1025.
25. Ota H, Katanosaka K, Murase S, Kashio M, Tominaga M, Mizumura K. TRPV1 and TRPV4 play pivotal roles in delayed onset muscle soreness. *PloS One*. 2013;8(6):e65751.
26. Ota H, Katanosaka K, Murase S, Furuyashiki T, Narumiya S, Mizumura K. EP2 receptor plays pivotal roles in generating mechanical hyperalgesia after lengthening contractions. *Scand J Med Sci Sports*. 2017;28:826-833.

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